

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Given

- **Training Mean Squared Error (MSE): 0.85**
- **Testing Mean Squared Error (MSE): 5.20**

The large difference between the training and testing errors indicates that the model performs well on the training data but poorly on unseen data. This is a classic sign of **overfitting**.

Possible Reasons for the Difference

1. **Overfitting**
 - The neural network has learned the training data, including its noise, instead of the underlying patterns.
 - As a result, it cannot generalize well to new data.
2. **Insufficient Training Data**
 - A small dataset may not capture all variations, causing the model to memorize rather than learn.
3. **Model Complexity**
 - Too many hidden layers or neurons increase the model's complexity and lead to overfitting.
4. **Poor Data Preprocessing**
 - Features may not be normalized or standardized.
 - Missing values or outliers can negatively affect performance.
5. **Train-Test Distribution Mismatch**
 - The training and testing datasets may have different data distributions.
6. **Lack of Regularization**

Without techniques such as L1/L2 regularization or Dropout, the model is more prone to overfitting.

Corrective Measures

- Increase the amount of training data or use data augmentation if applicable.
- Apply regularization techniques such as:
 - L1 Regularization
 - L2 Regularization
 - Dropout
- Reduce model complexity by decreasing the number of hidden layers or neurons.
- Normalize or standardize input features before training.
- Use Early Stopping to stop training when validation error starts increasing.
- Perform Cross-Validation to improve model generalization.
- Tune hyperparameters such as learning rate, batch size, and number of epochs.

Analysis

Metric	Training	Testing
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Mean Squared Error (MSE)	0.85	5.20
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- The training MSE (0.85) is very low, indicating that the model fits the training data well.
- The testing MSE (5.20) is much higher, showing poor performance on unseen data.
- The large gap between the two errors indicates high variance (overfitting).

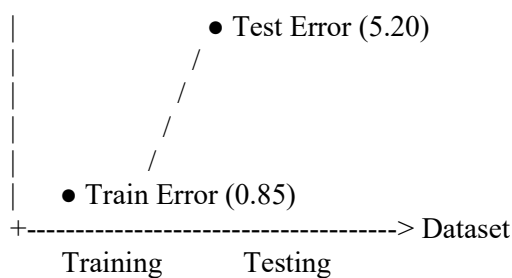
Conclusion

The neural network is overfitting the training dataset. To improve performance on unseen data, techniques such as regularization, early stopping, feature normalization, reducing model complexity, and increasing training data should be applied. These methods help the model generalize better and reduce the gap between training and testing errors.

Illustration (Training vs Testing Error)

MSE

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Interpretation: The large increase in testing error compared to training error indicates that the model has learned the training data too specifically and does not generalize well to new data.

Code of Conduct:

I K.P. Ajith Kumar/ 192425139 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided